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L1F22BSCS0335

Task 1

Purpose

Crypter's main purpose is to make a fun language based in cryptography

Style or Syntax

It is C++ like structure but cryptic keywords reversed, ~~unique rules~~ prefix and postfix for ~~variables~~ identifiers and variables

Reason

- It seemed fun
- Studied cryptography this semester in I.S.

Keywords

fi, esle, elihw, rot, nrutor

Operators

+, =, -

Punctuations

}, {

Task 2

1. Program Structure

Program $\rightarrow$ Statement-List

Statement-List $\rightarrow$ Statement Statement-List
 $\mid \epsilon$

2. Variable Declaration

Declaration $\rightarrow$ Data-Type ~~ID~~ Identifier

Data-Type $\rightarrow$ ~~int~~ | float | char | short
 $\mid$ long

3. Assignment

Assignment $\rightarrow$ Identifier
IDENTIFIER = Expression ;

Expression $\rightarrow$ Expression Operator Expression
 $\mid$ id $\mid$ num

Operator $\rightarrow$ + $\mid$ - $\mid$ * $\mid$ /

4. Conditional $\rightarrow$ ~~if~~ if Condition {

} Statement-List ϵ
 $\mid$ if Condition { Statement-List ϵ
else Statement-List ϵ

5. loop

Loop $\rightarrow$ while (condition) { Statement-List }

6. Input/Output

Input-Statement $\rightarrow$ scanf >> IDENTIFIER

Output-Statement $\rightarrow$ printf << Expression;

Task 3

10 rules list

Program $\rightarrow$ Statement-List

Statement-List $\rightarrow$ Statement Statement-List

Statement-List $\rightarrow$ ϵ

Statement $\rightarrow$ Declaration

Statement $\rightarrow$ Assignment

Statement $\rightarrow$ Conditional

Statement $\rightarrow$ Loop

Statement $\rightarrow$ Input-Statement

Statement $\rightarrow$ Output-Statement

Declaration $\rightarrow$ Data-Type IDENTIFIER;

Task 4

	First	Follow
Data_Type	{ tni, taolt, ranc gnirts, loob }	{ IDENTIFIER }
Statement	{ tni, taolt, ranc, gnirts, loob, IDENTIFIER fi, elihw, nic tuoc }	{ tni, taolt, ranc gnirts, loob , IDENTIFIER fi, elihw nic, tuoc, \$ }

Task 5

Expression $\rightarrow$ Expression + Expression
 | Expression - Expression
 | identifier | num

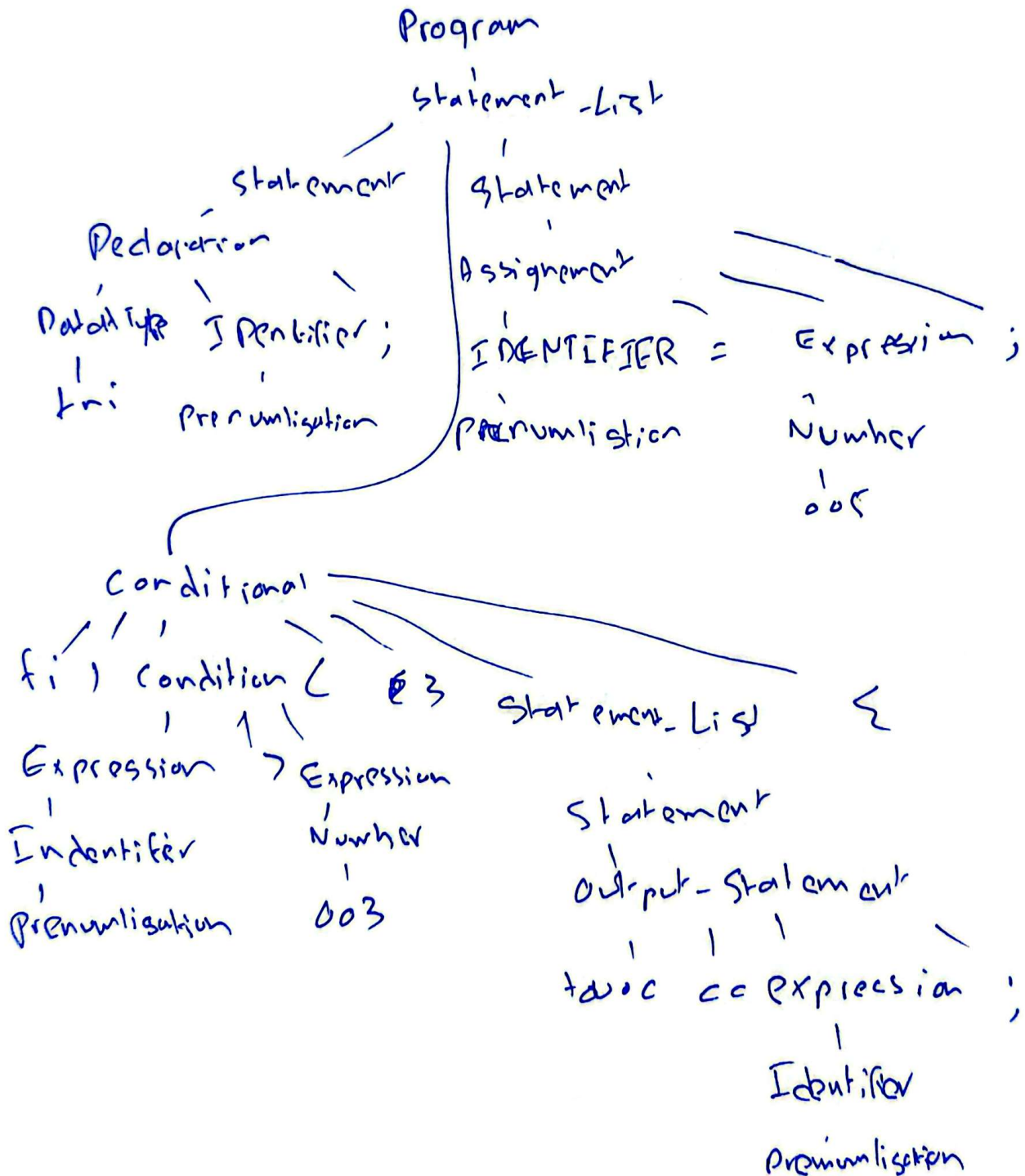
a + b - c which to do first? + or -
 we define + - as left associative

Task 6

fn: prenumlisation

prenumlisation = 005;

fi: prenumlisation ? 003 < @ 3 tu oc < c prenumlisation; {



Task 7

fn: presumlisation
presumlisation = 005;

Line of error

Line 1

rule

Declaration $\rightarrow$ Datatype Identifier;

Error

missing semi colon

```
fi { prextrate > 005 }  
    tue.cc xtrate;  
}
```

~~Line~~ Error Line

Line 1

Rule

fi) Condition (} Statement_List

Error

missing)